

Human-Computer Interaction

Johan F. HOORN and Jeff TANG

Lecture 01: Use and Context (continued)



Human-Computer Interaction

Last time we discussed:

The importance of HCI:

less redesign, cost reduction, avoiding disasters

History of HCI:

From batch jobs to social robots

From 'being an appendix to the machine' to 'the machine tending to the user's needs'

Visionaries and scruffy first prototypes are nothing to laugh at

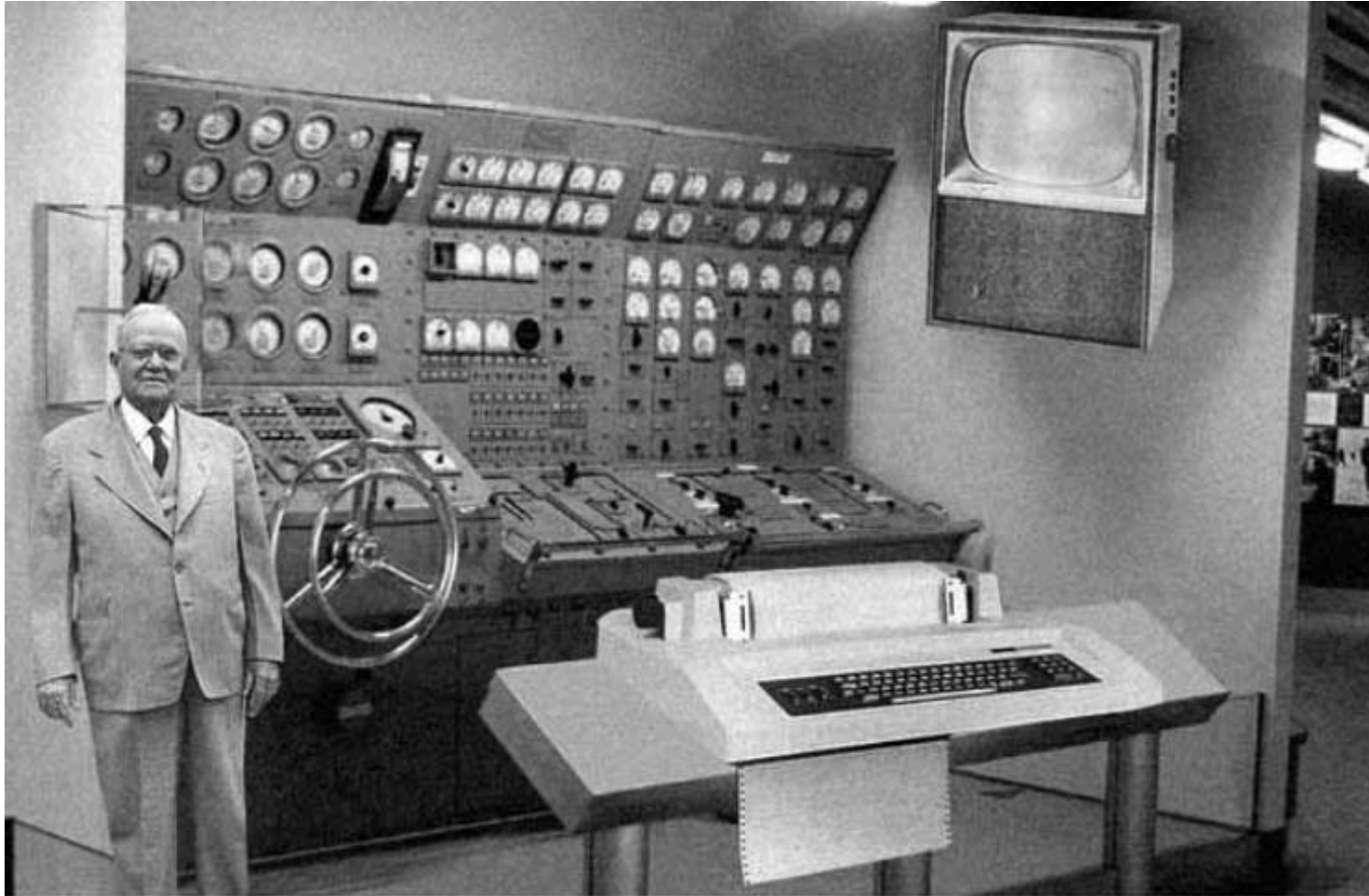
Take-home message:

Know use and context first before you start coding

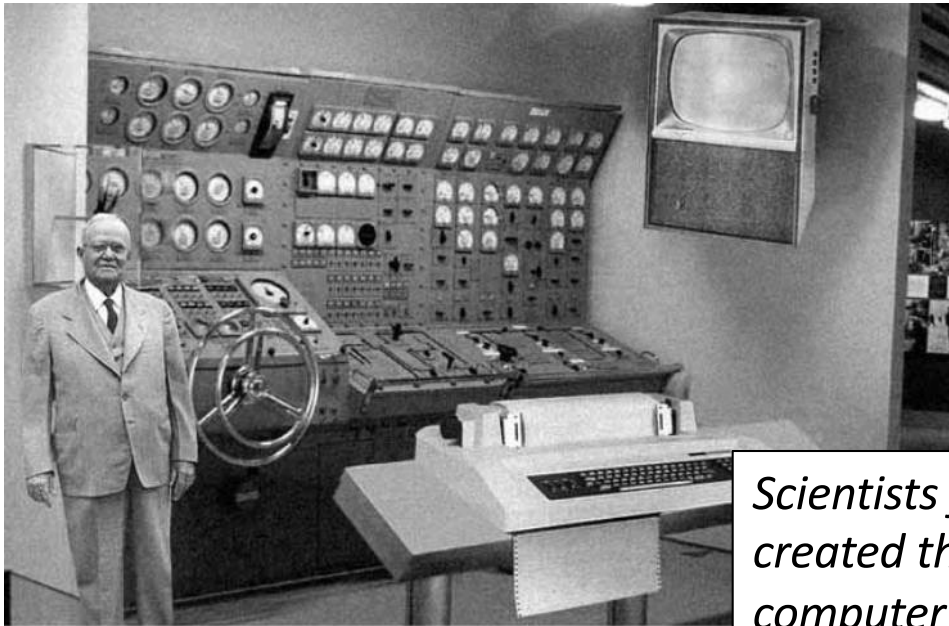
History of HCI

(continued)

Home computer according to RAND Corporation (1954)



Scientists from the RAND Corporation have created this model to illustrate how a "home computer" could look like in the year 2004. However the needed technology will not be economically feasible for the average home. Also the scientists readily admit that the computer will require not yet invented technology to actually work, but 50 years from now scientific progress is expected to solve these problems. With teletype interface and the Fortran language, the computer will be easy to use.

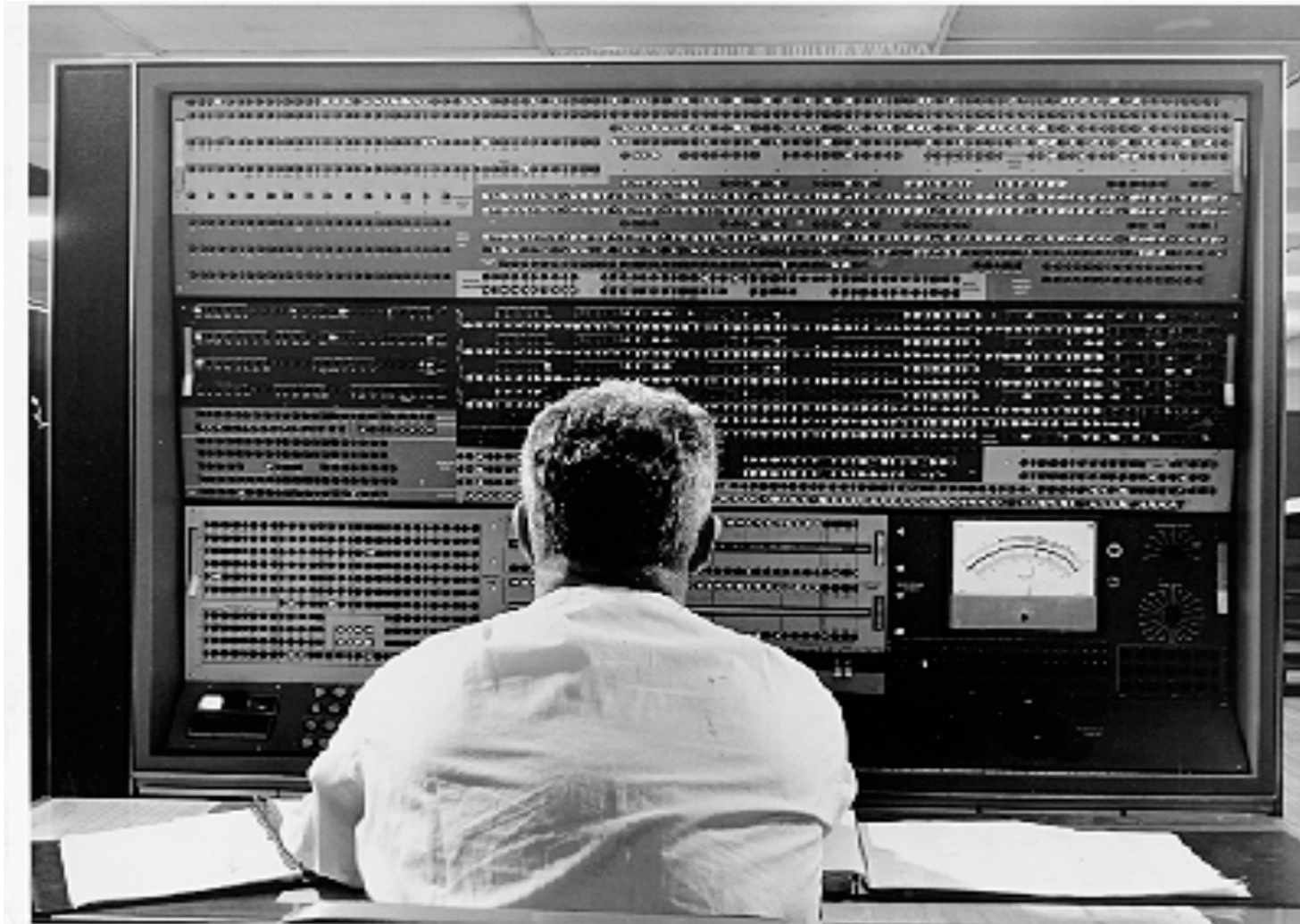


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History of HCI

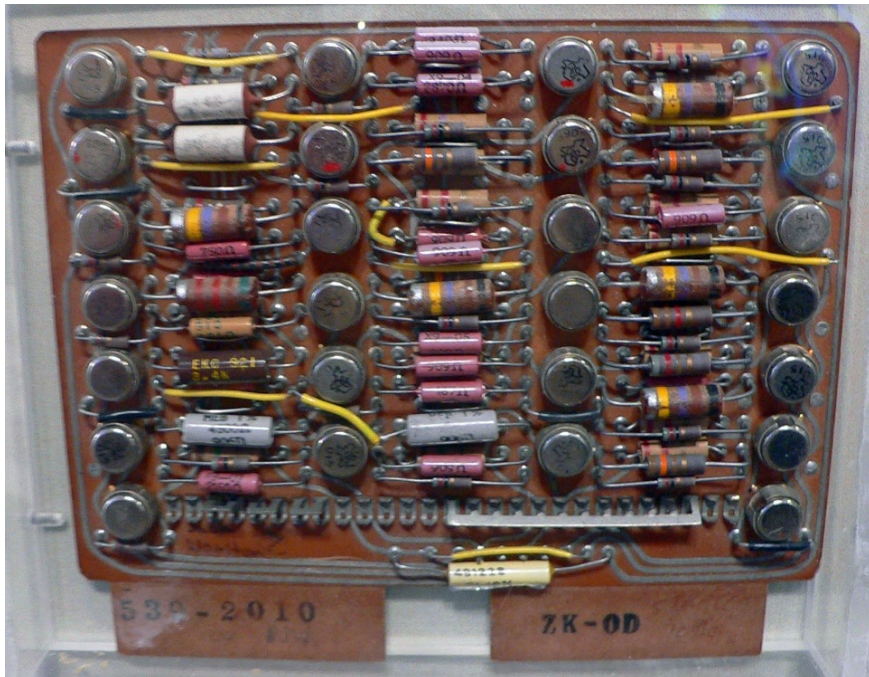
A close-up of the Stretch (1961) technical control panel



IBM 7030, transistorized supercomputer

History of HCI

Stretch Circuit board (double card)



Bradbury Science Museum, Los Alamos, New Mexico

- The 7030 CPU has 169,100 transistors, using 21 kW power
- Main memory = 16K-256K 64-bit binary words, in banks of 16K
- Memory temperature regulated in an oil bath to keep performance stable
- Accepted dedicated program languages as well as FORTRAN

History of HCI

Significant Advances 1960 - 1980

❑ Mid 60's

- Computers too expensive for single person



CTSS: Compatible Time Sharing System (MIT)

- ### ❑ Time sharing gave the *illusion* of one being on one's own computer
- Dramatically increased accessibility of computers
 - Led to immediate need for Human-Computer Interaction
 - afforded interactive systems/languages vs batch “jobs”
 - community as a whole communicated through computers

History of HCI

Interaction starts to happen

❑ Ergonomics (1956-63)

- Console ergonomics
- Job control languages
- CTSS, MAC, JOSS, BASIC
- Transistors and core stores

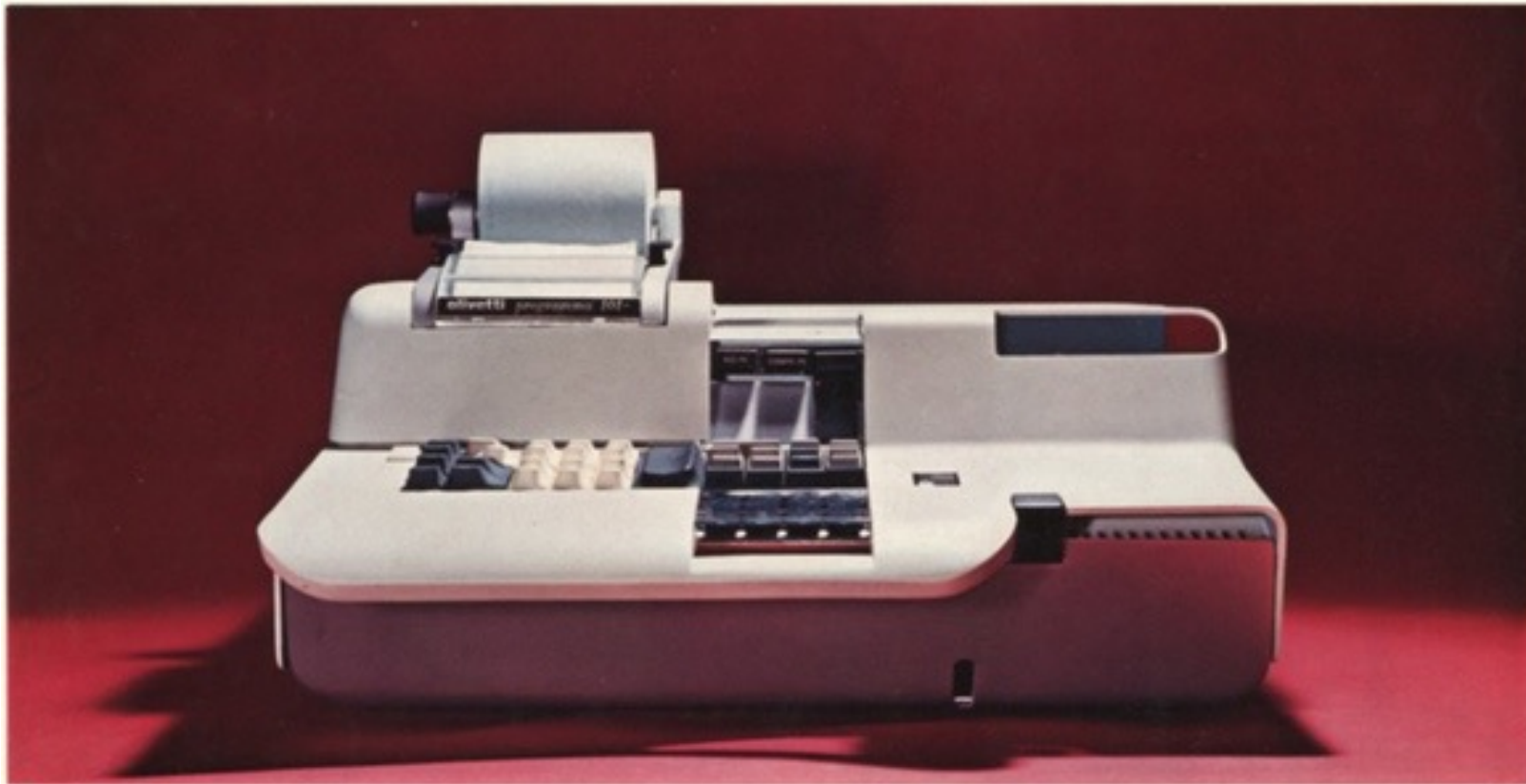


❑ Interactive Experience (1964-71)

- Interactive terminals, command shells, speech synthesis
- TSS360, APL/360, Large Scale ICs

History of HCI

Significant Advances 1960 - 1980



Programma 101 (Olivetti, 1965).
First PC ever

History of HCI

Computers are getting cheaper

❑ HCI Design Rules (late 1960s-70s)

- Personal computing
- Dialog rules
- Videotex services

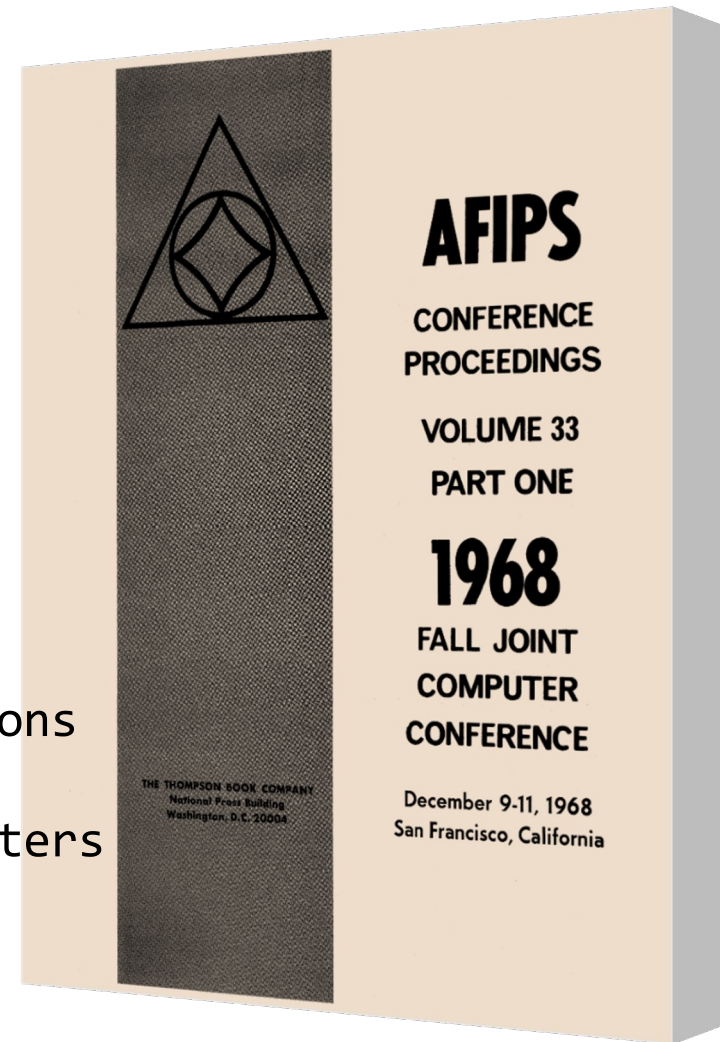


Minitel, French videotex system

History of HCI

AFIPS Fall Joint Computing Conference (1968)

- ☐ Document Processing
 - modern word processing
 - outline processing
 - hypermedia
- ☐ Input / Output
 - the mouse and one-handed corded keyboard
 - high resolution displays
 - multiple windows
 - specially designed furniture
- ☐ Shared work
 - shared files and personal annotations
 - electronic messaging
 - shared displays with multiple pointers
 - audio/video conferencing
 - ideas of an Internet
- ☐ User testing, training



History of HCI

The user becomes important (1980-87)

❑ ACM SIGCHI (1982)



“The scope of [SIGCHI](#) consists of the study of the human-computer interaction process and includes research and development efforts leading to the design and evaluation of user interfaces. The focus of SIGCHI is on how people communicate and interact with computer systems.”

❑ HCI journals

- International Journal of Man-Machine Studies (1969)
- International Journal of Human-Computer Studies (IJHCS)
- ACM Transactions on Computer-Human Interaction (TOCHI)



History of HCI

Joseph C. R. Licklider (1960)

❑ Famous paper: Man-Computer Symbiosis

“The hope is that, in not too many years, human brains and computing machines will be coupled together very tightly and that the resulting partnership will think as no human brain has ever thought and process data in a way not approached by the information-handling machines we know today.”

<http://groups.csail.mit.edu/medg/people/psz/Licklider.html>

Man-Computer Symbiosis

J. C. R. Licklider

IRE Transactions on Human Factors in Electronics,
volume HFE-1, pages 4-11, March 1960

Summary

Man-computer symbiosis is an expected development in cooperative interaction between men and electronic computers. It will involve very close coupling between the human and the electronic members of the partnership. The main aims are 1) to let computers facilitate formulative thinking as they now facilitate the solution of formulated problems, and 2) to enable men and computers to cooperate in making decisions and controlling complex situations without inflexible dependence on predetermined programs. In the anticipated symbiotic partnership, men will set the goals, formulate the hypotheses, determine the criteria, and perform the evaluations. Computing machines will do the routinizable work that must be done to prepare the way for insights and decisions in technical and scientific thinking. Preliminary analyses indicate that the symbiotic partnership will perform intellectual operations much more effectively than man alone can perform them. Prerequisites for the achievement of the effective, cooperative association include developments in computer time sharing, in memory components, in memory organization, in programming languages, and in input and output equipment.

! Introduction

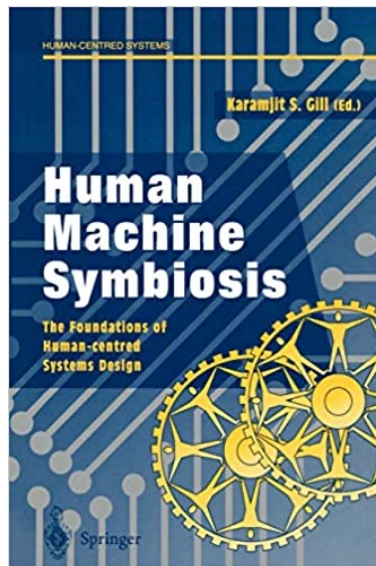
! Symbiosis

*Fig tree is pollinated only by the insect *Blastophaga grossorum*. The larva of the insect lives in the ovary of the fig tree, and there it gets its food. The tree and the insect are thus heavily*

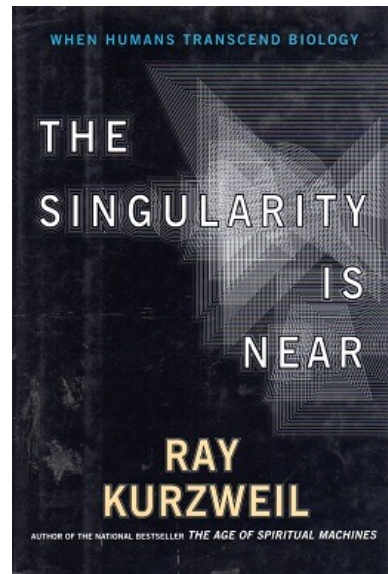
History of HCI

Joseph C. R. Licklider (1960)

❑ Famous paper: Man-Computer Symbiosis



1996

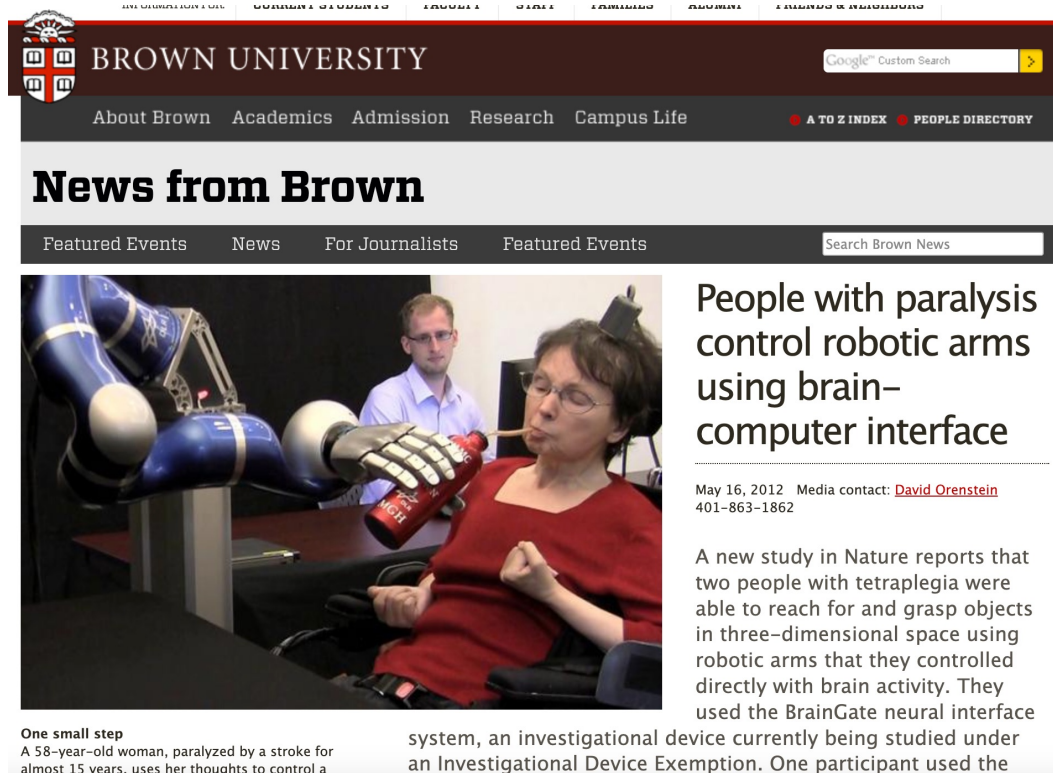


2005



2018

Brain-computer interface



The screenshot shows the Brown University website's news section. The header includes the university's name and a search bar. Below the header, there's a navigation bar with links like 'About Brown', 'Academics', 'Admission', 'Research', and 'Campus Life'. The main section is titled 'News from Brown' and includes a sub-navigation bar with 'Featured Events', 'News', 'For Journalists', and 'Featured Events'. A search bar for 'Search Brown News' is also present. The featured article is titled 'People with paralysis control robotic arms using brain-computer interface'. It includes a photo of a woman in a red shirt sitting at a desk with a robotic arm. The text of the article describes a study in Nature where two people with tetraplegia used the BrainGate neural interface to control robotic arms. The date is May 16, 2012, and the media contact is David Orenstein.

People with paralysis control robotic arms using brain-computer interface

May 16, 2012 Media contact: [David Orenstein](#)
401-863-1862

A new study in Nature reports that two people with tetraplegia were able to reach for and grasp objects in three-dimensional space using robotic arms that they controlled directly with brain activity. They used the BrainGate neural interface system, an investigational device currently being studied under an Investigational Device Exemption. One participant used the

One small step
A 58-year-old woman, paralyzed by a stroke for almost 15 years, uses her thoughts to control a

Close to being cyborgs

- pacemakers
- cochlear implants
- artificial lenses
- exoskeletons
- blades - lower leg
- chips under the skin
- metal joints (knee, hip)
- kidney dialysis
- Intensive Care unit



- International Journal of Human-Computer Studies (IJHCS)



International Journal of Human-Computer
Studies

Volume 181, January 2024, 103142



The media inequality, uncanny mountain, and the singularity is far from near: Iwaa and Sophia robot versus a real human being

Johan F. Hoorn^a  , Ivy S. Huang^b

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<https://doi.org/10.1016/j.ijhcs.2023.103142> 

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Highlights

- Human-likeness is about intrinsic rather than extrinsic qualities of a humanoid robot.
- Human-likeness is not higher for human-looking robots compared to non-human looking robots.
- Human and non-human looking robots are not experienced differently (i.e. involvement, eeriness, task-related experiences).

<https://www.sciencedirect.com/science/article/pii/S1071581923001519?via%3Dihub>

History of HCI

Joseph C. R. Licklider (1960)

❑ Famous paper: Man-Computer Symbiosis

Short term goals

- time sharing of computers among many users
- electronic I/O for the display and communication of symbolic and pictorial information
- interactive real time system for information processing and programming
- large scale information storage and retrieval

Intermediate

- facilitation of human cooperation in the design and programming of large systems
- speech recognition, hand-printed character recognition, and light-pen editing

Long term vision

- natural language understanding (syntax, semantics, pragmatics)
- speech recognition of arbitrary computer users
- heuristic programming



Hypertext Editing System (HES)
with light pen (1969)

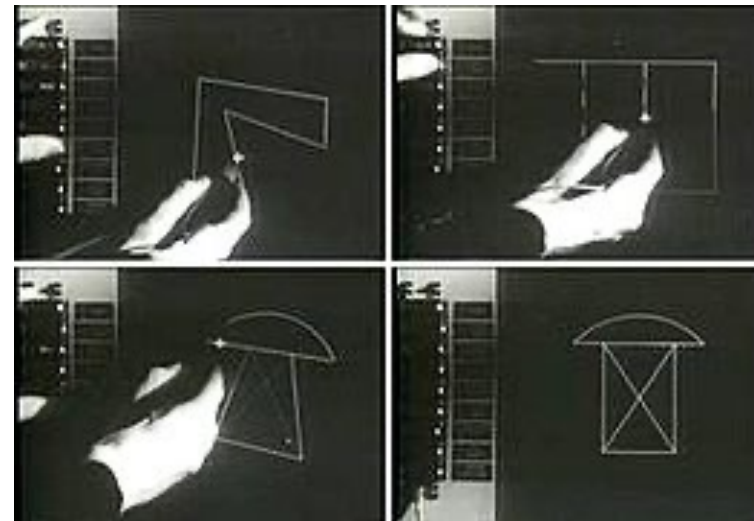
History of HCI

Ivan Sutherland's SketchPad (1963)

□ Drawing tool

- Hierarchical structures defined pictures and sub-pictures
- Object-oriented programming: master picture with instances
- Constraints: specify details which the system maintains through changes
- Icons: small pictures that represented more complex items
- Copying: both pictures and constraints
- Input techniques: efficient use of light pen

Sounds familiar?



Graphical user interface

History of HCI

Ivan Sutherland

<https://uxdesign.cc/how-apple-makes-soft-ui-the-future-9f3ac69eea6f>



Steve Jobs?
Never heard of

History of HCI

Ivan Sutherland

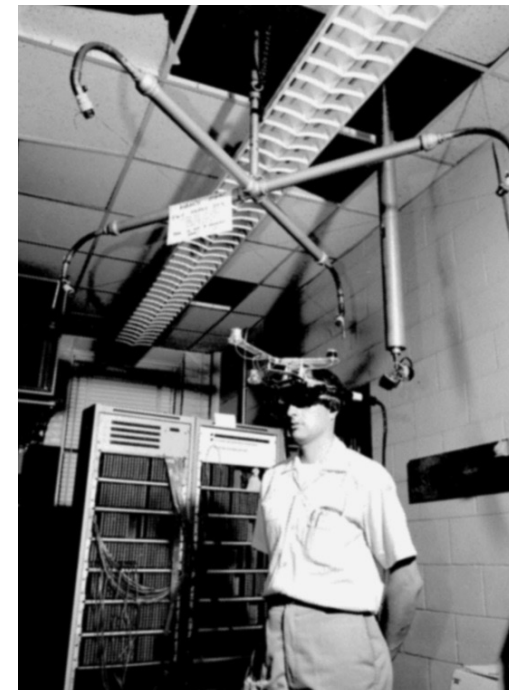
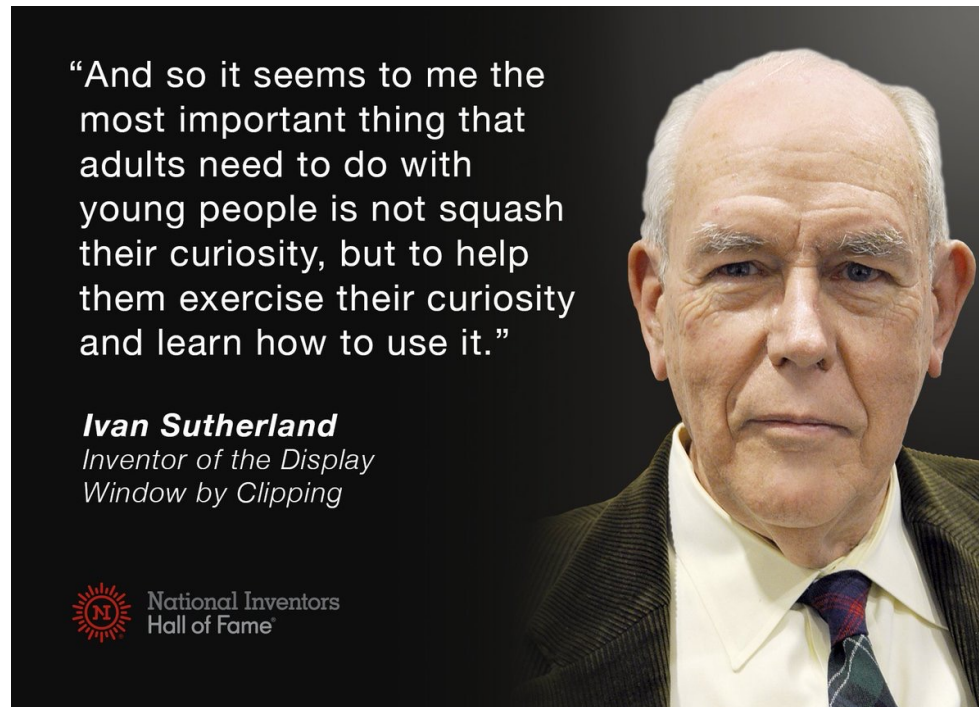
□ Parallel developments in hardware

- “Low-cost” graphics terminals
- Input devices such as data tablets (1964)
- Display processors capable of real-time manipulation of images (1968)

Gamers, pay tribute!



3D computer graphics and head-mounted displays



History of HCI

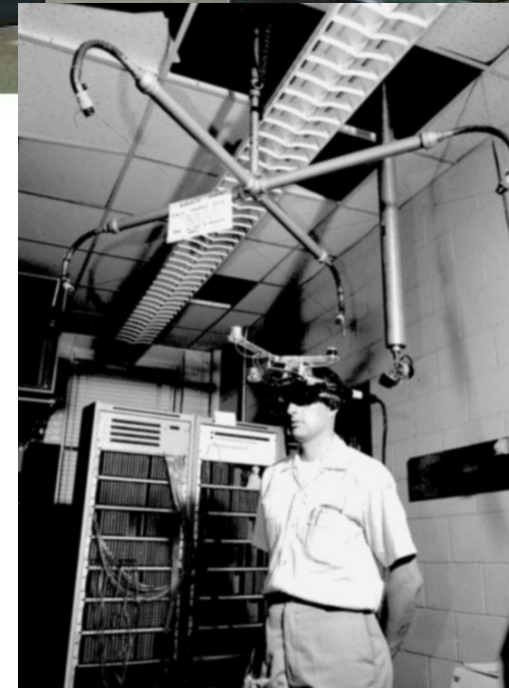
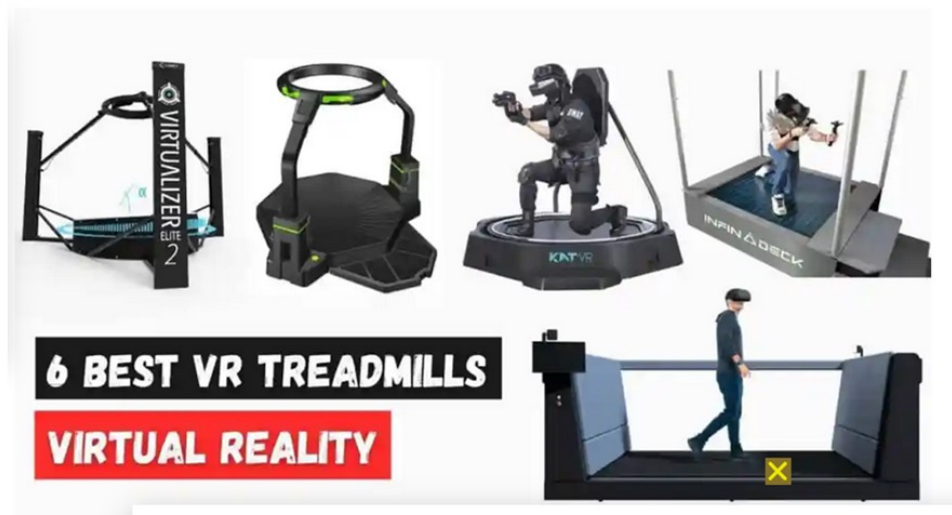
Ivan Sutherland

Gamers, pay tribute!



6 Best VR Treadmills 2021: Omnidirectional Virtual Reality Haptic Treadmill

Virtual Reality Gadgets, Electronic Gadgets, Tech Gadgets, Tech News, VR Games / By techhunters / March 13, 2021



History of HCI

Ivan Sutherland

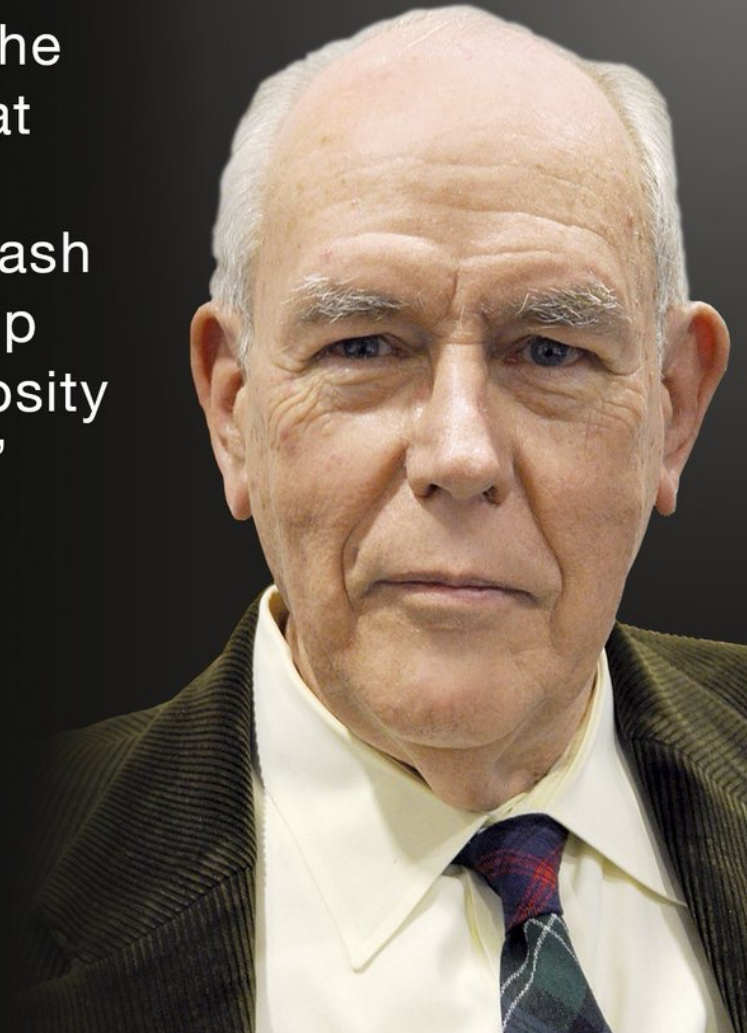
“And so it seems to me the most important thing that adults need to do with young people is not squash their curiosity, but to help them exercise their curiosity and learn how to use it.”

Ivan Sutherland

*Inventor of the Display
Window by Clipping*



National Inventors
Hall of Fame®

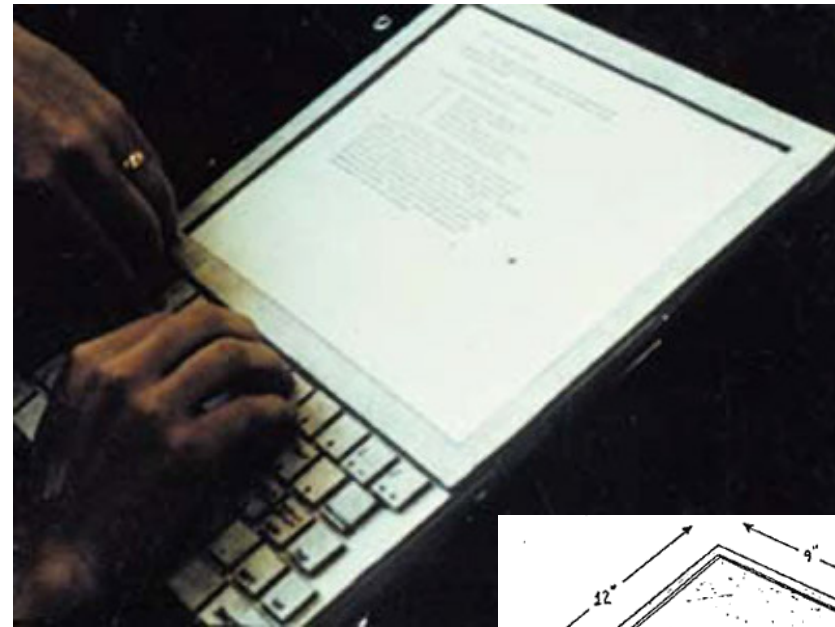
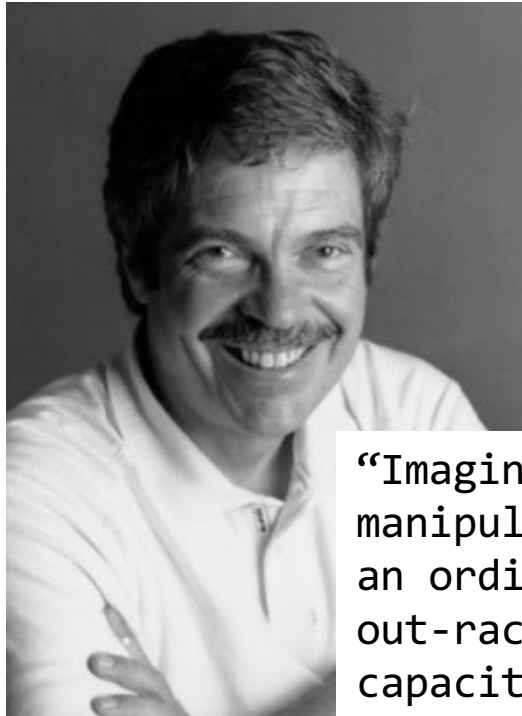


History of HCI

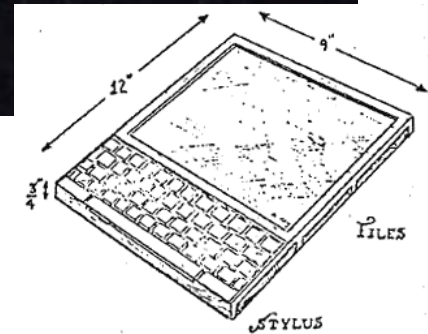
Alan Kay (1969)

- Personal computing

Dynabook vision of a notebook



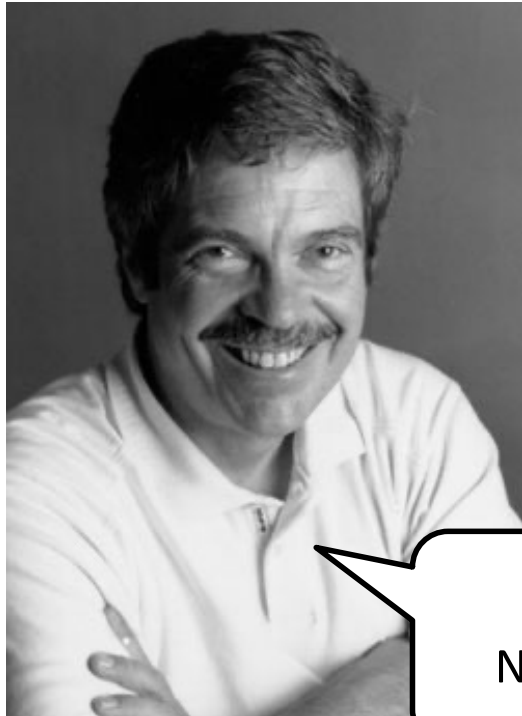
Cardboard prototype!



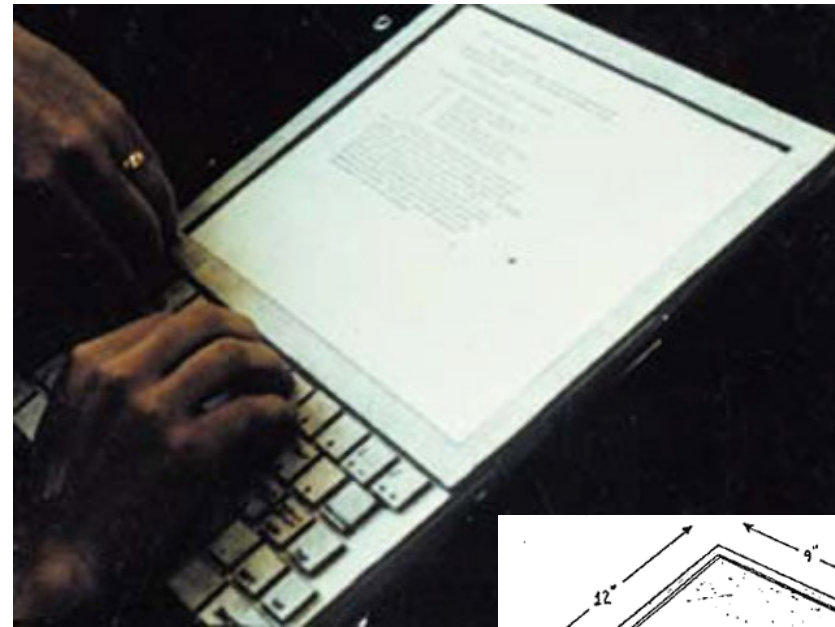
“Imagine having your own self-contained knowledge manipulator in a portable package the size and shape of an ordinary notebook. Suppose it had enough power to out-race your senses of sight and hearing, enough capacity to store for later retrieval thousands of page-equivalents of reference materials, poems, letters, recipes, records, drawings, animations, musical scores...”

History of HCI

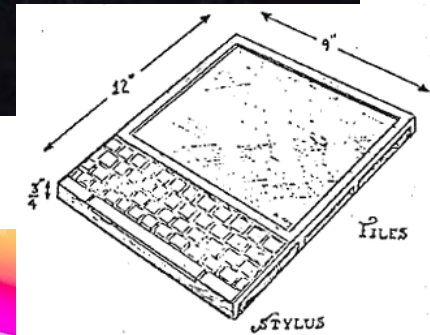
Alan Kay (1969)



Steve Jobs?
Never heard of



Cardboard prototype!



Apple MacBook Air (M1, 2020)

History of HCI

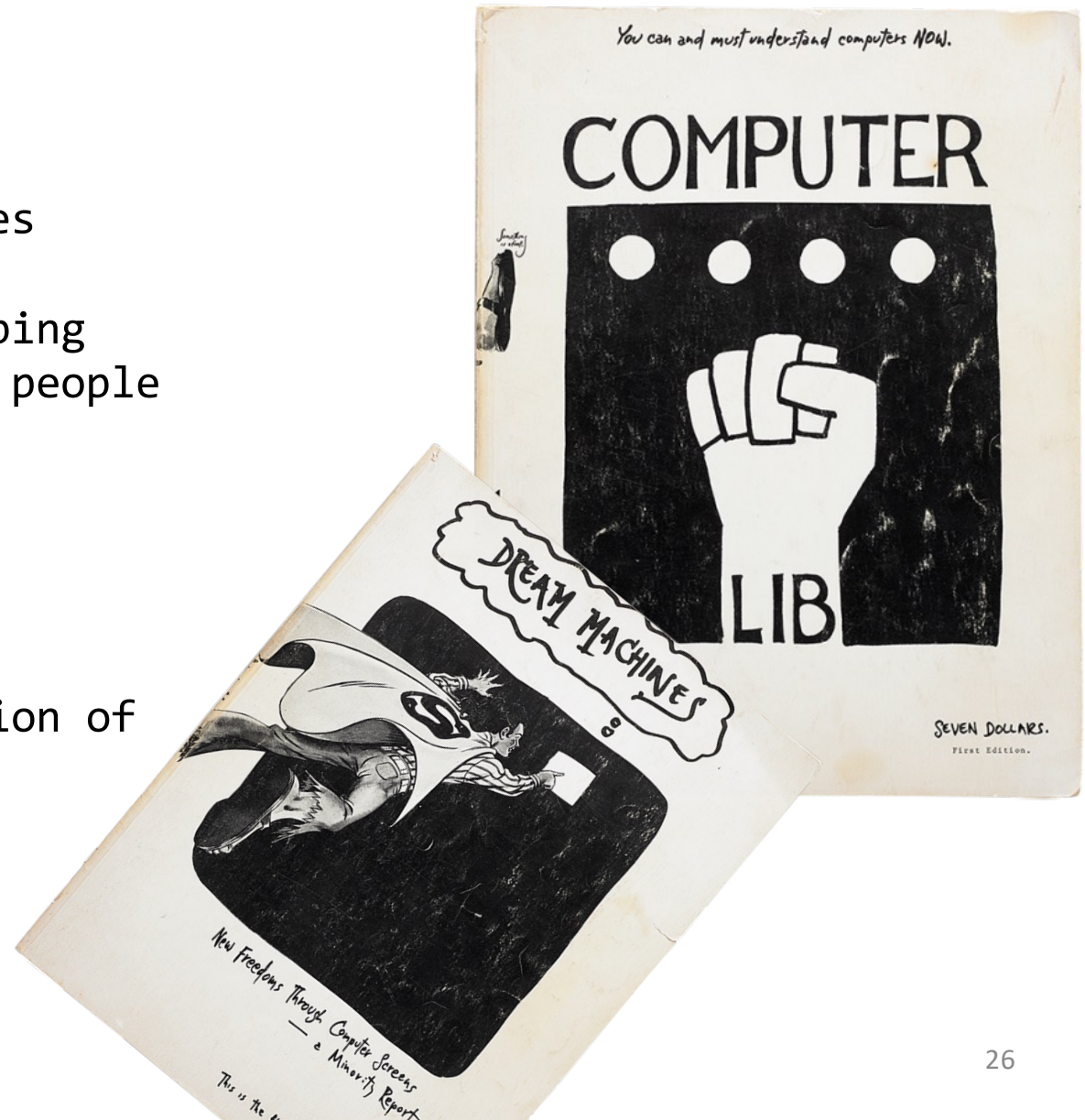
Ted Nelson (1974)

- Personal computing

Computer Lib/Dream Machines

2-front cover book describing
what computers can do for people
(instead of business!)

- Deep computer literacy
- Counter culture
- Against the centralization of computers (i.e. IBM)



History of HCI

Xerox PARC (1973)

- Personal computing (GUI!)
- Alto Computer, a personal workstation with local processor, bit-mapped display, mouse
- Modern graphical interfaces with text and drawing editing, electronic mail, windows, menus, scrollbars, mouse selection, etc.
- Local area network (Ethernet), using shared resources



Display screen has portrait orientation

History of HCI

Quiz item!!

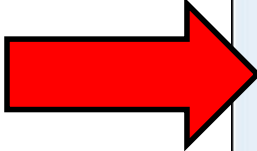
- Raise your hand if you know who it is.
Have you ever heard of:
 - Ed McCreight
 - Steve Jobs
 - Chuck Thacker
 - Steve Wozniak
 - Butler Lampson
 - Bob Sproull
 - Dave Boggs

History of HCI

Xerox PARC (1973)

Xerox PARC Research Center Looks Back on 40 Years of Invention

The Alto Computer, 1973



In 1972, Xerox decided to produce a personal computer for research purposes. This became the Alto computer, the result of a joint effort by Ed McCreight, Chuck Thacker, Butler Lampson, Bob Sproull and Dave Boggs, who were attempting to make a device that was small enough to fit in an office comfortably yet powerful enough to support a reliable, high-quality operating system and graphics display. Its GUI featured windows and icons. A few years later, Steve Jobs and Steve Wozniak borrowed some of these ideas and started Apple Computer.

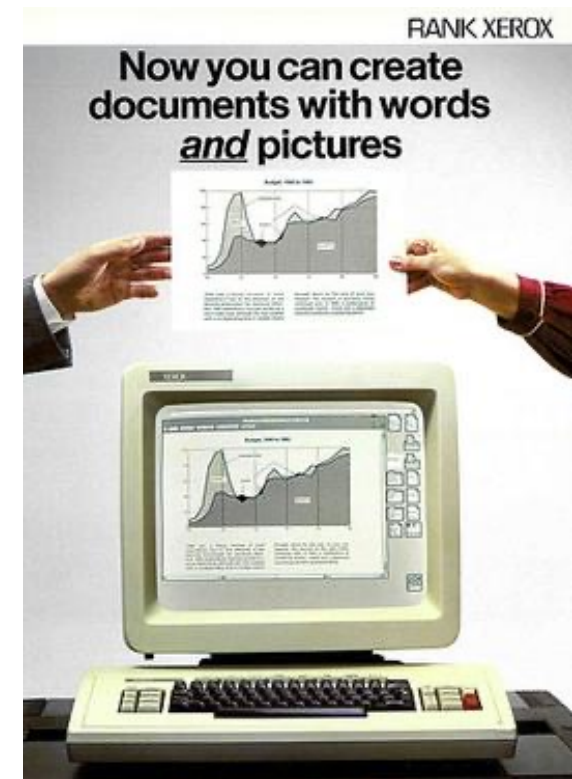


<https://sphs73reunion.org/webpages/zremtech.html>

History of HCI

Xerox Star 8010 (1981)

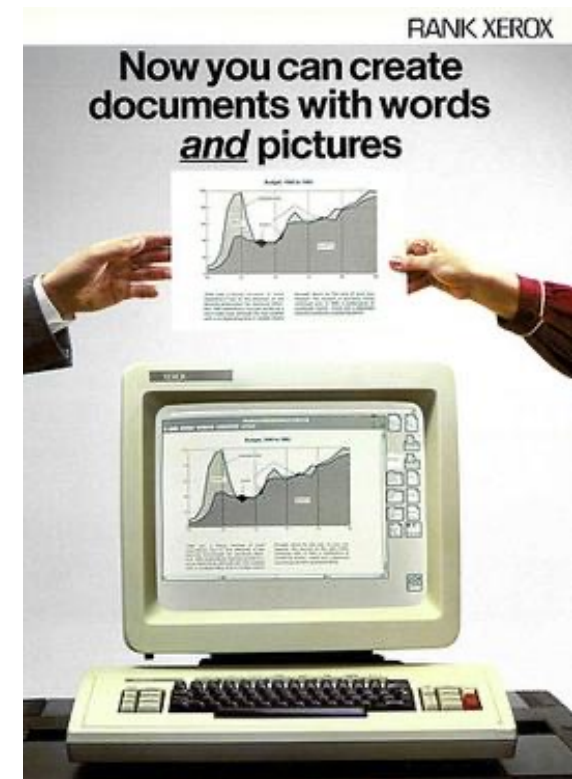
- Personal computing (commercialized)
 - First commercial personal computer for “business professionals”
 - First comprehensive GUI, using many ideas developed at Xerox PARC
 - familiar conceptual model (simulated desktop)
 - promoted recognizing/pointing rather than remembering/typing
 - property sheets to specify appearance/behavior of objects
 - what you see is what you get (WYSIWYG)
 - small set of generic commands that could be used throughout the system
 - high degree of consistency and simplicity
 - modeless interaction
 - limited amount of user customization



History of HCI

Xerox Star 8010 (1981) (continued)

- Personal computing (commercialized)
 - First system based upon usability engineering
 - inspired design
 - extensive paper prototyping and usage analysis
 - usability testing with potential users
 - iterative refinement of interface
 - Commercial failure
 - cost (\$15,000) – IBM had just announced a less expensive machine!
 - limited functionality – e.g., no spreadsheet
 - closed architecture – 3rd party vendors could not add applications
 - perceived as slow – but really fast!
 - slavish adherence to direct manipulation



History of HCI

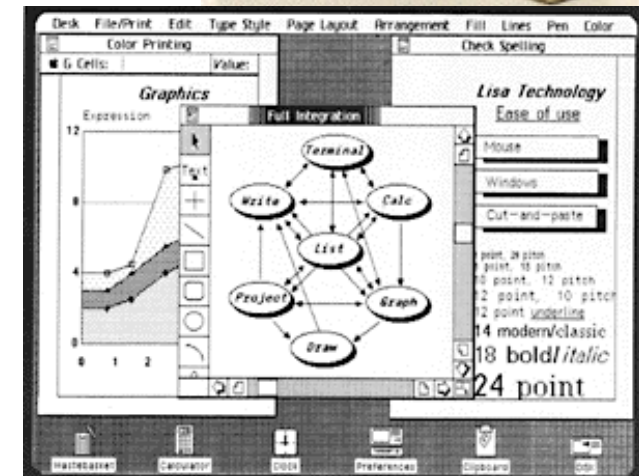
The user becomes important (1980-87)

□ User natural – systemic principles

- Human protocol
- Windows and graphical interfaces
- PCs have power and storage of mainframes
- Graphics and speech processing

□ Commercial machines: Apple Lisa (1983)

- based upon many ideas in the Star
- predecessor of Macintosh
- somewhat cheaper (\$10,000)
- commercial failure as well

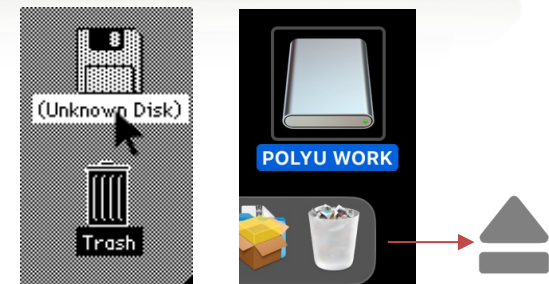


History of HCI

The user becomes important (1980-87)

❑ Commercial machines: Apple Macintosh (1984)

- “old ideas” but well done!
- succeeded because: sharp pricing (\$2,500)
- did not need to trail blaze
- corrected mistakes of Lisa; “mature” ideas
- market now ready
- developer’s toolkit encouraged 3rd party non-Apple software
- interface guidelines encouraged consistency between applications
- domination in desktop publishing because of affordable laser printer and excellent graphics



However:
Pull to Trash to eject?

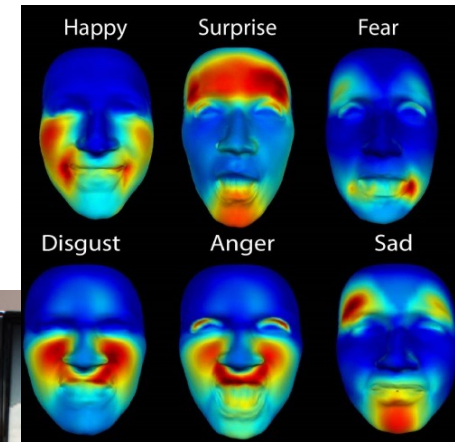
History of HCI

User takes center stage (1988-current)

❑ User similar - Automated design

- Integrated multi-modal systems
- Emotion detection
- Ubiquitous computing

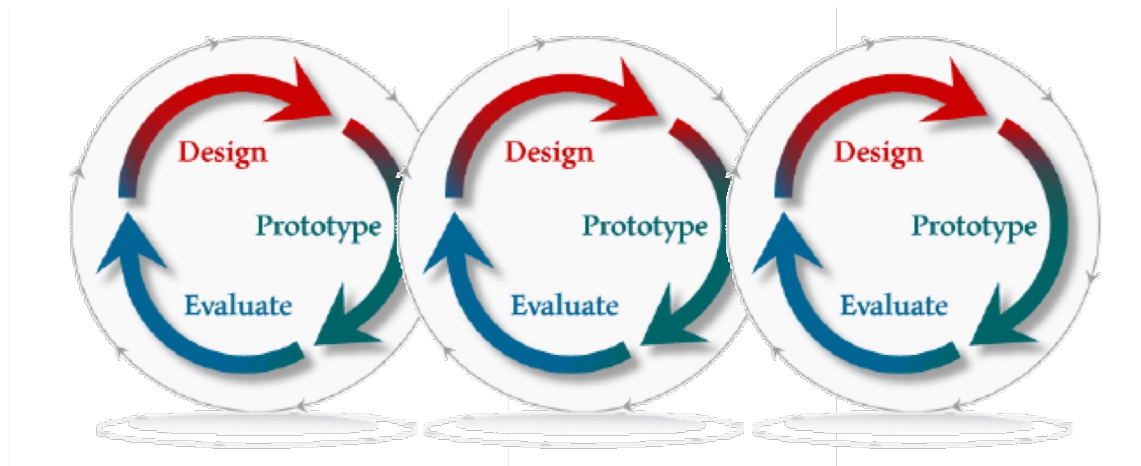
Sentiment-analysis system
(Hoque & Picard, 2011)



History of HCI

Lessons learned

- ❑ Before: Computer cost more than people
Today: People cost far more than computers do
- ❑ Applications can now afford to trade off processing power for human usability
- ❑ New ideas often don't make it into products straight away
- ❑ Pioneer systems developed innovative designs, but often commercially unviable
- ❑ Settler systems incorporated (many years later) well-researched (HCI!!) designs



Human-Computer Interaction

Johan F. HOORN and Jeff TANG

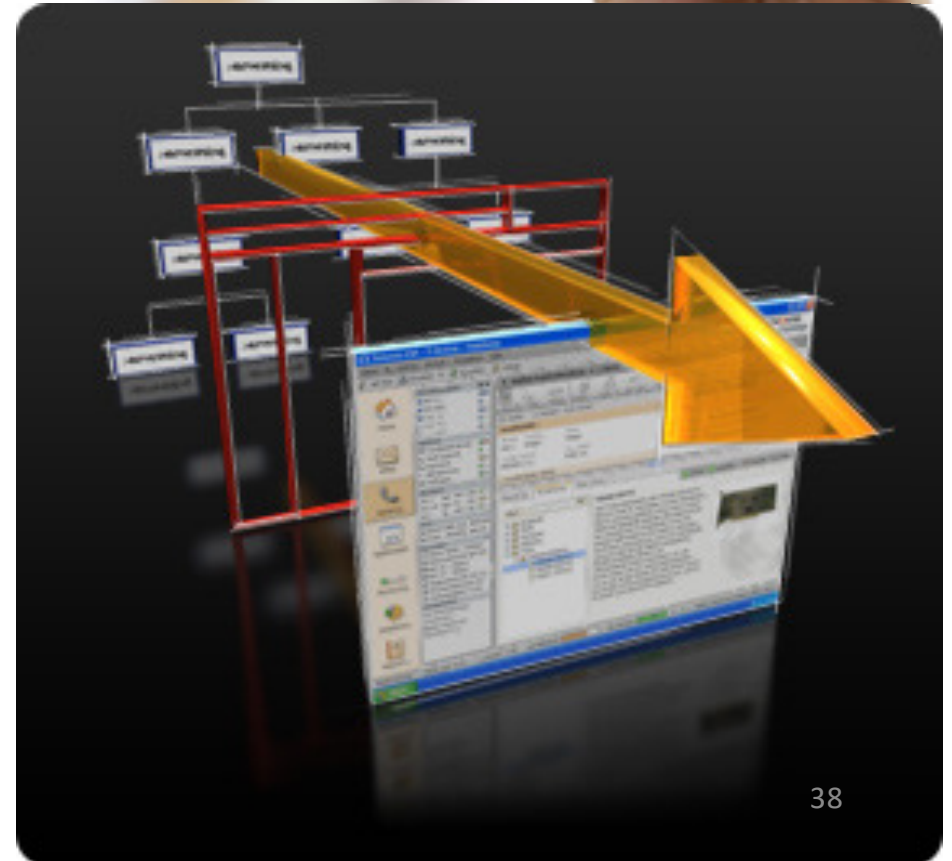
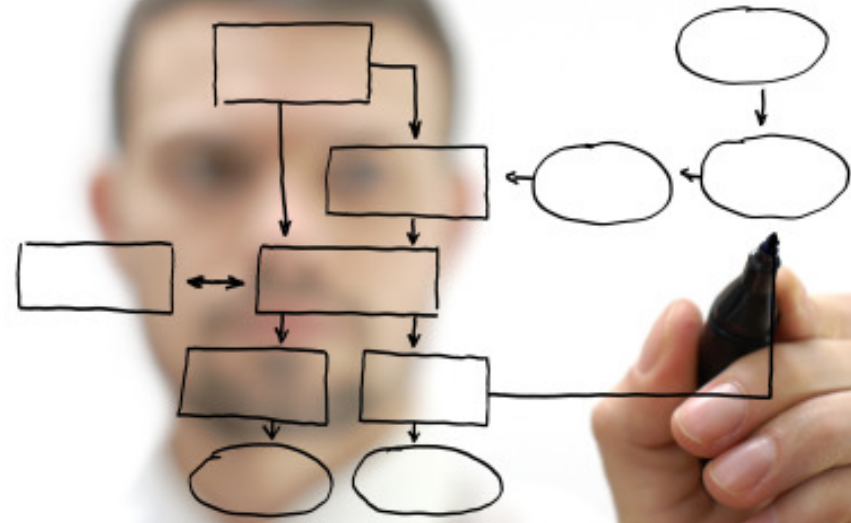
Lecture 01: Use and Context (continued)



HCI Today

Multidisciplinary

- ❑ Computer science
- ❑ Artificial intelligence
- ❑ Engineering
- ❑ Cognitive psychology
- ❑ Ergonomics, human factors
- ❑ Sociology
- ❑ Communication
- ❑ Anthropology
- ❑ Linguistics
- ❑ Industrial design
- ❑ &c

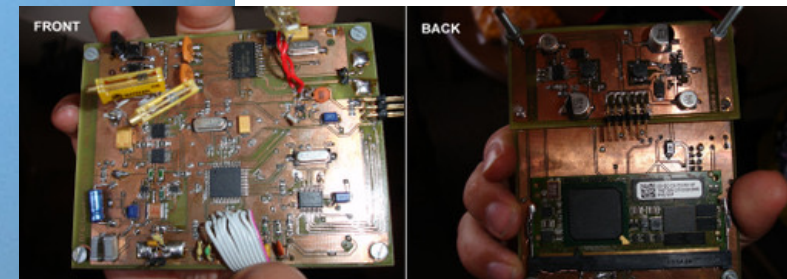


US Department of Homeland Security wants driver's licenses with RFID tag that can be read from a distance to use for patrol at the US-Mexican border



❑ Computer science

Melanie Rieback, Ubisec researcher who invented the first RFID virus at VU University Amsterdam



Melanie developed RFID Guardian, a personal firewall to protect privacy. It's an ordinary circuit board with two antennae and on-board processors. It intercepts signals from RFID readers and like a software firewall, it won't let those signals reach your RFID unless you want them to -- for example, if you're passing through customs

❑ Artificial intelligence

TRANSPORTATION | UBER | RIDE-SHARING

Uber reportedly thinks its self-driving car killed someone because it 'decided' not to swerve

The car's sensors saw her, but may have flagged the detection as a 'false positive'

By Sean O'Kane | @sokane1 | May 7, 2018, 2:41pm EDT

f 🐦  SHARE



❑ Engineering

- Douglas Engelbart (1962) “Augmenting Human Intellect: A Conceptual Framework”
- In 1968, workstation with a mouse, links across documents, chorded keyboard



Logitech MX Revolution Mouse



❑ Cognitive psychology

Log In to Your Account

Account Name

Password

☒ **Remember Me**

Secure Sign In

[Forgot password?](#) [Forgot account name?](#)

Click on the check box next to "Remember Me" before you log in so that your browser will remember your log in credentials

For instance, memory overload causes security issues:

- Passwords too simple ('password')
- Related to user self (birth date)
- Repeating same password everywhere
- Forgetting password (and account name)

→ Programming routines to make up for user error

❑ Ergonomics/human factors



Pacific Writing Instruments: The PenAgain



Maltron Single Handed Ergonomic Keyboard



"It's called Ergonomics."

<http://www.ds-ergonomics.com/images/picCartoonErgonomics.jpg>



"Ergonomics"
of a virtual
reality
gait-
recognition
suit in
human-
factors
research

❏ Sociology

How people
establish social
networks

Friday, November 30th, 2018

Keyword:

Search

networksasia



NETWORKING

MOBILITY

STORAGE

SECURITY

APPLICATIONS

UCS

DATA CENTER

Wireless Mobility

How Samsung uses social media analytics to understand customers and guide strategy

By Tom Macaulay | Friday, October 27, 2017 - 10:10

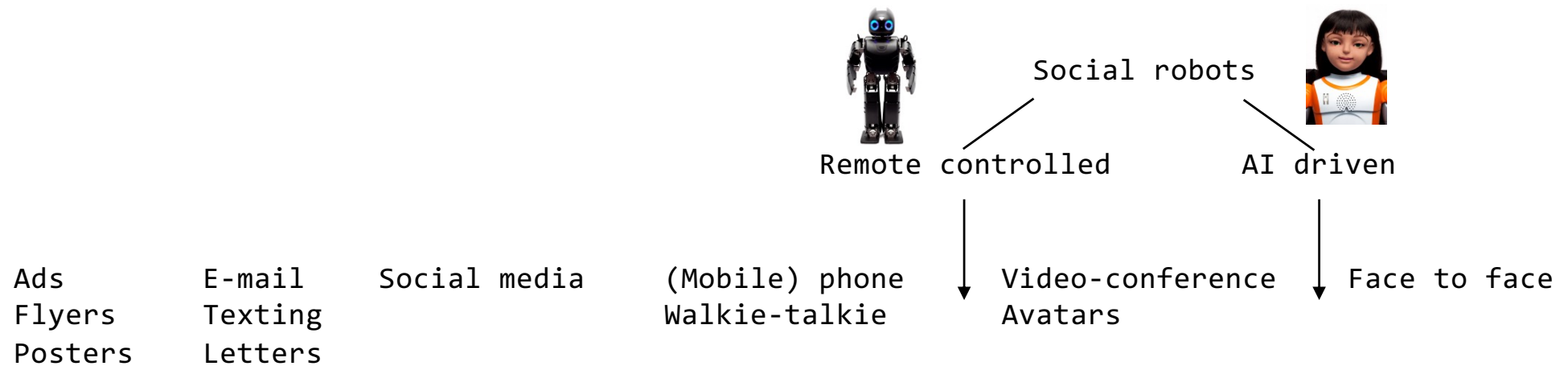
EmailSharePrint



Samsung is in the midst of a fierce battle with Apple for smartphone supremacy. One of the ways in which the South Korean company looks

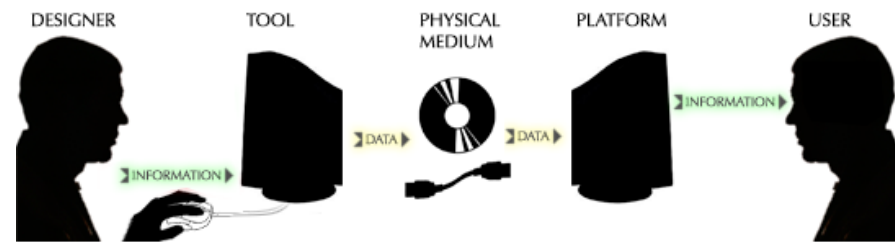
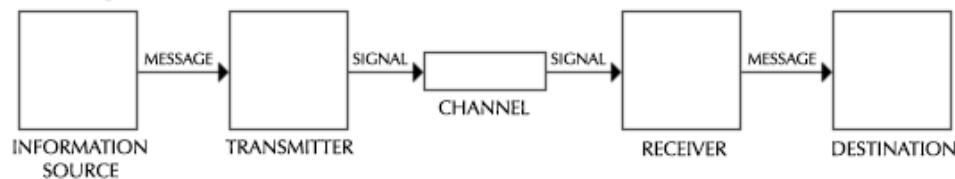
to get an edge over its nemesis across the Pacific Ocean is by mining insights from its consumers through their social media activity.

❏ (Computer-mediated) Communication



More cues to face to face communication in the medium →
supposedly lead to better communication and interpersonal relationship

Shannon's general communication model*



Analogous model for tools and platforms

❑ Anthropology

Calling-line identification

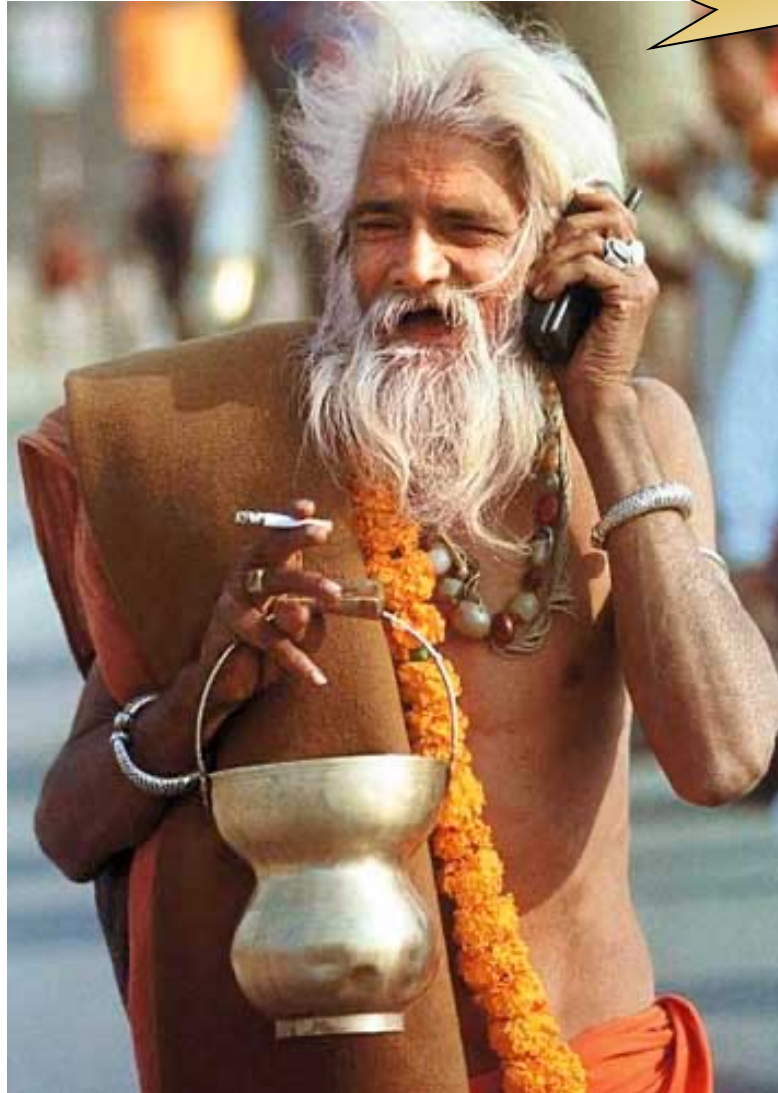
Non-Western cultures may dislike this

Hi Bert



❑ Anthropology

Hi Sandjai



Konkka, K. (2003). Indian needs. Cultural end-user research in Mumbai. In C. Lindholm, T. Keinonen, & H. Kiljander (Eds.), *Mobile Usability. How Nokia Changed the Face of the Mobile Phone* (pp. 97-111). New York, London: McGraw-Hill.

bluetooth burka



submitted by [maxkiosko](#) at 07.02.2008 via <http://seamless.sigtronica.org/projects.html>

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Hype Yourself
or get **dissed** trying.



GET HYPEDISS FOR
YOUR BLOG OR WEBSITE

6 **HYPE IT!**

0 **DISS IT!**

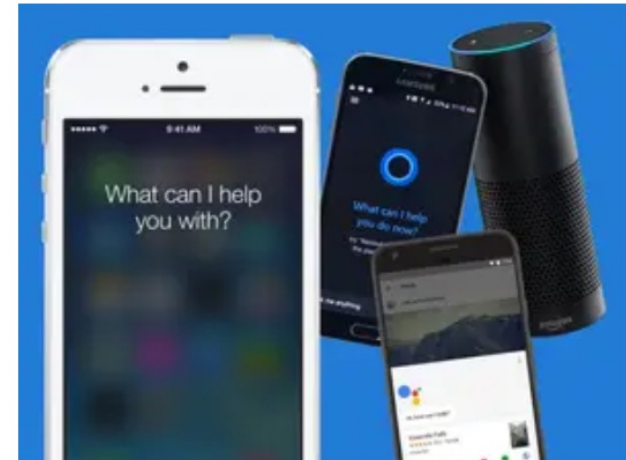
During Seamless: Computational Couture, the "Charming Burka" was introduced to the public. This burka sends out a self-defined picture of the wearing person via Bluetooth. A model presented the Charming Burka on the catwalk while the audience could receive the portrait of the model on their mobile phones.

Siri, Alexa, Google Assistant, Cortana

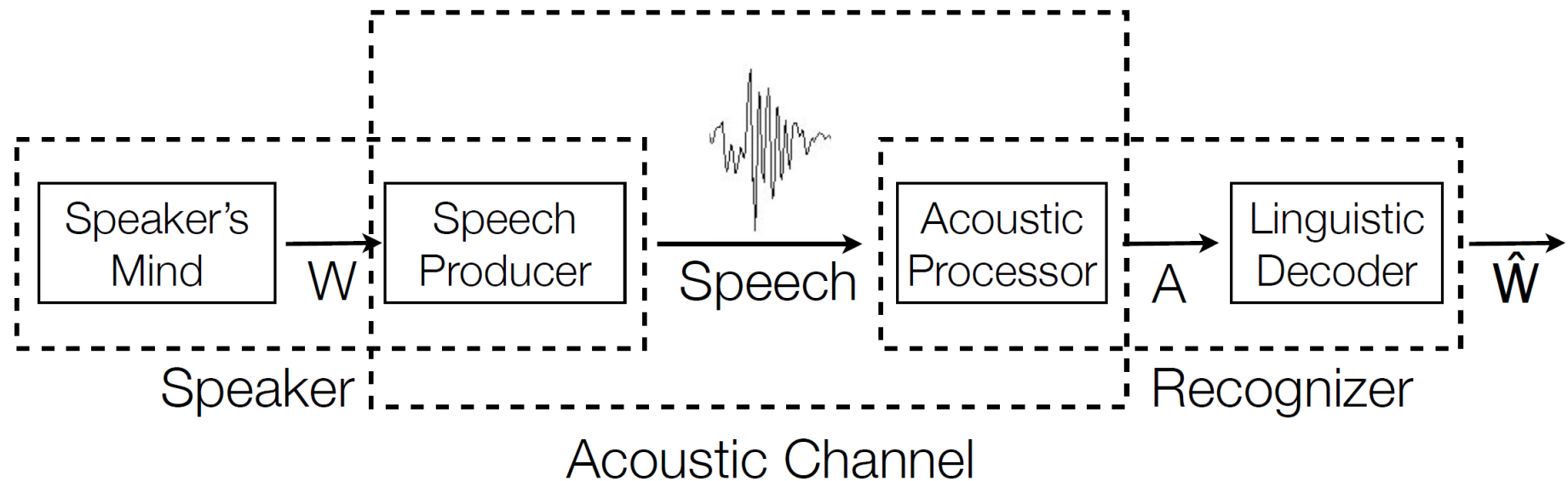
□ Linguistics

Two forms of speech recognition:

- Converts spoken words to machine-readable input (text)
- Identifies the speaker according to acoustic input (voice verification)
- Large Language Models (i.e. GPT)



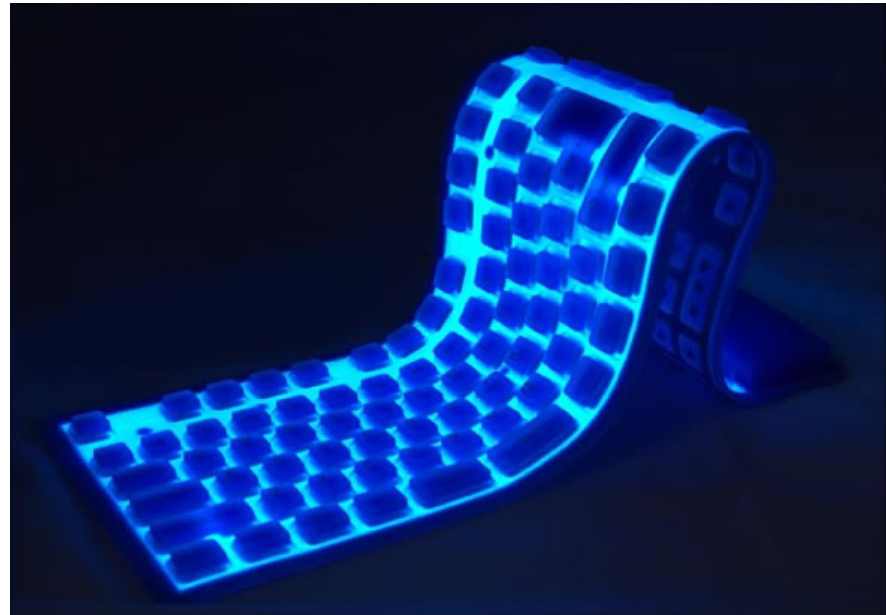
Business Insider/Yu Han



❑ Industrial design



The new Duraflex Comfort keyboard is resistant to water, alcohol, and other disinfectants including lighter cleaning products. Stash it in your bag.



The Bendi Board lights up in the dark

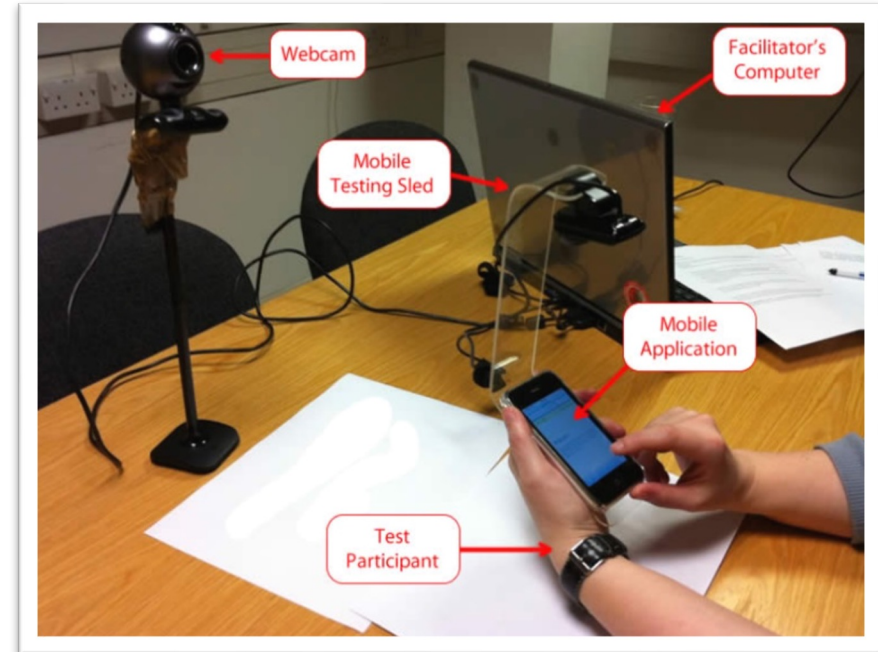
Designing Interactive Systems

❑ Understanding users and their tasks

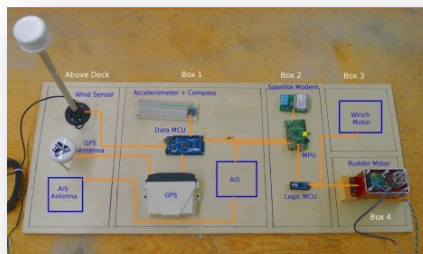
- Task-centered system design
 - Develop task examples
 - Evaluate designs through a task-centered walk-through

❑ Designing *with* the user

- User-centered design and prototyping
 - Methods for designing with the user
 - Low and medium fidelity prototyping
- Evaluating systems and interfaces *with* users
 - The role of evaluation in design
 - Observe people using systems to detect (interface) problems



User test: walk-through



Mock-ups

Low-fidelity
paper prototyping



Designing Interactive Systems

□ Designing visual interfaces

- Design of everyday things
 - What makes visual design work?
- Beyond screen design
 - Representations and metaphors
- Graphical screen design
 - The placement of interface components on a screen



□ Principles of design

- Design principles, guidelines, and usability heuristics
 - Using guidelines to design and discover usability problems



$$T = a + b \log_2 \left(2 \frac{D}{W} \right)$$

Time

Distance

Coefficients

Width

The wider an interface widget and the closer it is to the pointer, the easier to hit it

Fitts' Law

Designing Interactive Systems

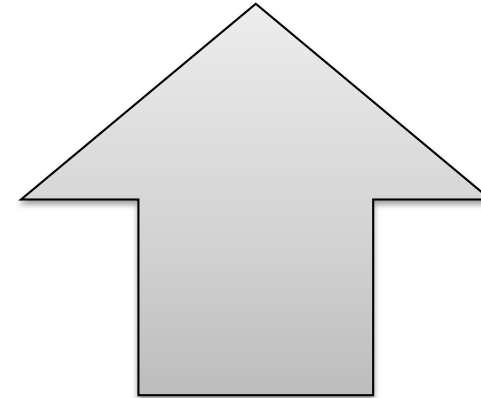
❑ Let humans focus on:

- Sensing of low level stimuli
- Pattern recognition
- Inductive reasoning
- Multiple strategies
- Adapting
- Creating

❑ While computers help with:

- Counting and measuring
- Accurate storage and recall
- Rapid and consistent responses
- Data processing
- Calculation
- Repetitive actions

Pattern
recognition,
artificial
creativity



Currently,
computers
are moving
up

Virtual Tutor Summary



In this course, we teach you why HCI is crucial in system design and how to make sure the user becomes an integral part of your thinking

HCI is not only important to create a nice user experience, to avoid costly redesign, or to sell more products or services, there are shocking examples where bad design caused human disaster (e.g., Harrisburg or American Airlines 965)

By definition, HCI concerns the design, implementation, and evaluation of interactive computing systems for human use and provides input language for the user, an output language for the machine, and a protocol for interaction

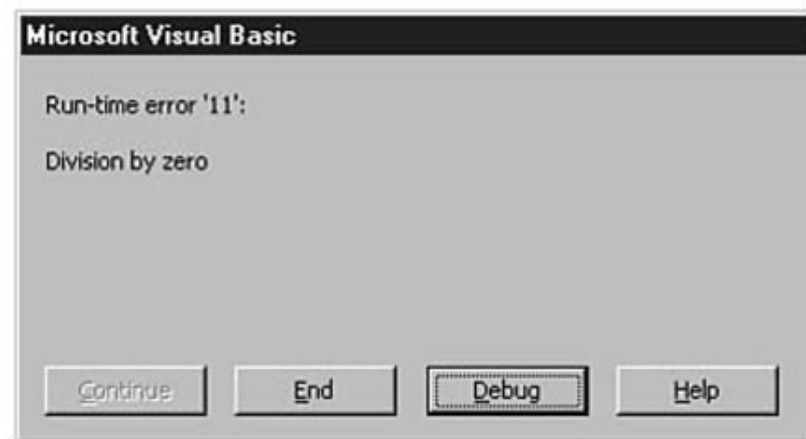
Virtual Tutor Summary



In historical context, the focus shifts from humans servicing computer systems to systems servicing humans

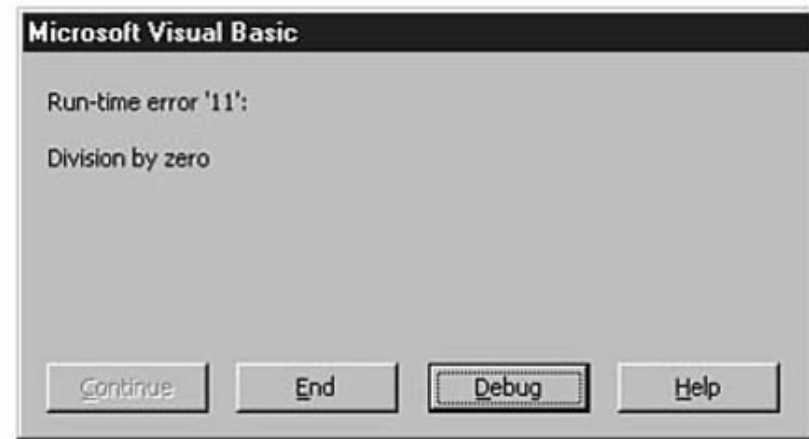
Various disciplines should collaborate if a computer system wants to be fit for human use: Computer science, social science and psychology, ergonomics, and interaction design

If not, poor HCI design makes even the toughest coder look stupid



Just a little joke

```
Private Sub Command1_Click()  
  
h = human  
c = computer  
x = coder  
  
If h * c ≥ 1 Then  
    Print "Way to go!"  
Else  
    x = x / 0 ' divide coder by zero  
    Print "x = "; x  
End If  
  
End Sub
```



Exam Sample Item

Category “Insight”

How is Fitts’ Law related to the fat-finger syndrome?

- A Stress worsens performance
- B Fitts reckons with the width ratio between pointer and interface widget
- C The logarithmic distribution for fat fingers differs from, for instance, mouse pointers



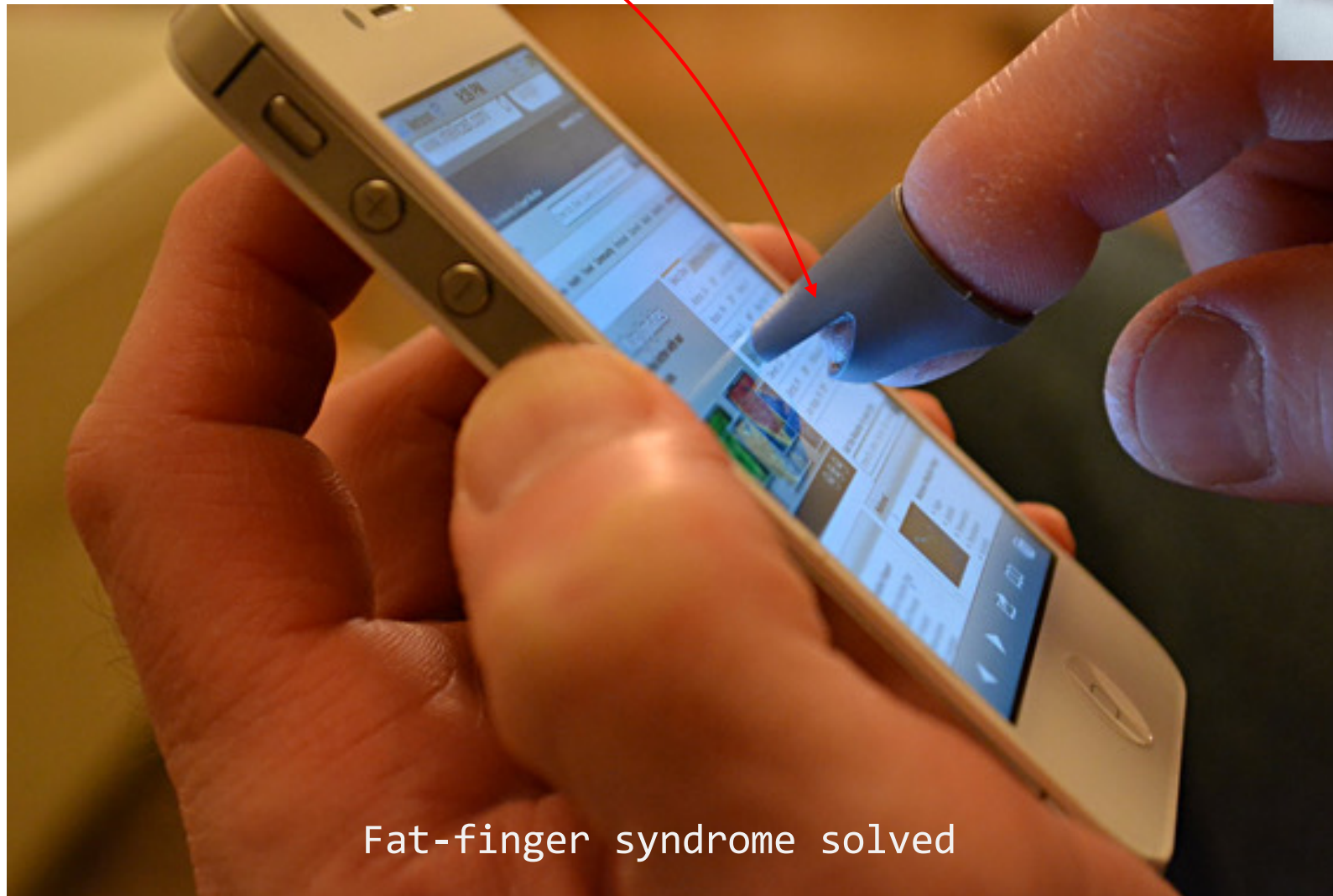
A and B are correct

Fitts: The wider an interface widget and the closer it is to the pointer, the easier to hit it - particularly when the pointer is wide as well (fat finger). So it’s about the pointer-to-target width ratio, not target width alone. And yes, people make more errors under stress

Importance of HCI

❑ Ergonomic design

Like a thimble.
Solved through
creative design



Fat-finger syndrome solved